

GRANBY, QC, Canada

WMO: 710367

Lat: 45.3727N

Lon: 72.7743W

Elev: 86

StdP: 100.30

Time Zone: -5.00 (NAE)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.2	(c) -22.4	(d) -29.1	(e) 0.3	(f) -24.9	(g) -26.6	(h) 0.3	(i) -21.8	(j) 9.0	(k) -5.3	(l) 8.1	(m) -7.8	(n) 1.7	(o) 220	(p) 0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.4	(c) 29.8	(d) 21.9	(e) 28.3	(f) 21.1	(g) 26.9	(h) 20.2	(i) 23.5	(j) 27.7	(k) 22.5	(l) 26.5	(m) 21.6	(n) 25.2	(o) 2.7	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.1	17.0	25.7	21.2	16.0	24.7	20.4	15.2	23.9	70.3	27.9	66.7	26.6	63.6	25.0	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.1	(b) 6.0	(c) 5.2	(e) -27.8	(f) 32.5	(g) 4.5	(h) 1.3	(i) -31.1	(j) 33.4	(k) -33.7	(l) 34.1	(m) -36.2	(n) 34.9	(o) -39.5	(p) 35.8		
(a) 7.1	(b) 6.0	(c) 5.2	(e) -28.3	(f) 25.1	(g) 4.6	(h) 1.3	(i) -31.6	(j) 26.1	(k) -34.4	(l) 26.8	(m) -37.0	(n) 27.6	(o) -40.3	(p) 28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-8.8	-7.7	-2.3	5.6	13.4	17.8	21.0	19.7	15.6	9.1	1.9	-4.3	(6)
(7)		DBStd	11.66	8.15	6.66	6.56	4.67	4.56	3.69	2.82	3.02	4.21	4.72	5.35	6.50	(7)
(8)		HDD10.0	2395	583	496	384	146	18	0	0	0	4	74	247	443	(8)
(9)		HDD18.3	4424	842	729	641	382	163	54	8	22	103	286	494	701	(9)
(10)		CDD10.0	1237	0	0	3	14	124	233	340	299	171	48	4	0	(10)
(11)		CDD18.3	223	0	0	0	1	11	37	89	63	21	1	0	0	(11)
(12)		CDH23.3	1663	0	0	1	8	118	307	675	396	153	6	0	0	(12)
(13)		CDH26.7	350	0	0	0	1	22	68	163	73	23	0	0	0	(13)
(14)	Wind	WSAvg	2.0	2.5	2.3	2.4	2.6	2.1	1.8	1.5	1.5	1.9	2.3	2.2	(14)	
(15)	Precipitation	PrecAvg	1208	87	71	81	91	102	110	124	118	110	113	95	106	(15)
(16)		PrecMax	1464	165	172	124	178	212	194	176	214	205	197	134	214	(16)
(17)		PrecMin	991	39	37	36	19	50	50	53	49	42	50	40	52	(17)
(18)		PrecStd	114	26	26	19	40	37	37	29	33	39	45	23	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	10.0	19.6	24.6	29.3	30.9	32.2	30.7	29.3	24.3	18.2	12.1	(19)
(20)			MCWB	10.5	7.3	14.5	15.0	18.5	21.9	24.0	23.3	22.6	18.0	14.4	9.6	(20)
(21)		2%	DB	8.0	6.0	12.1	20.0	26.5	28.6	30.0	28.6	26.8	21.5	15.2	8.6	(21)
(22)			MCWB	6.1	3.6	7.8	11.6	17.9	20.9	21.9	21.5	20.4	16.4	12.1	6.8	(22)
(23)		5%	DB	4.8	3.8	9.1	16.7	24.1	26.7	28.5	27.0	25.2	19.2	12.2	6.2	(23)
(24)			MCWB	3.2	2.0	5.9	10.8	16.6	19.3	21.7	20.6	19.8	15.1	9.5	5.0	(24)
(25)		10%	DB	2.4	2.1	6.6	13.8	21.7	24.8	27.0	25.6	23.1	16.6	9.5	4.0	(25)
(26)			MCWB	1.2	0.9	4.2	8.4	15.3	18.4	20.8	20.3	18.8	13.2	7.3	2.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	8.5	14.8	16.2	21.7	23.7	25.4	24.7	23.6	19.4	15.2	10.1	(27)
(28)			MCD B	11.7	9.8	19.1	22.5	26.3	28.9	30.5	28.7	27.9	22.3	17.7	11.3	(28)
(29)		2%	WB	6.5	3.9	8.7	13.7	19.7	22.3	23.7	22.8	22.2	17.8	12.7	7.1	(29)
(30)			MCD B	7.5	5.6	10.8	18.2	24.3	26.6	27.6	26.9	25.8	21.0	15.3	8.2	(30)
(31)		5%	WB	3.3	2.2	6.1	11.2	18.1	21.1	22.7	21.9	20.8	15.7	10.1	5.1	(31)
(32)			MCD B	4.4	3.4	8.2	15.5	21.9	24.8	26.6	25.3	23.9	18.3	12.1	6.1	(32)
(33)		10%	WB	1.2	0.8	4.2	9.2	16.4	19.8	21.8	21.1	19.4	13.8	7.8	2.9	(33)
(34)			MCD B	2.3	2.0	6.7	13.1	20.4	23.4	25.4	24.4	22.6	16.2	9.7	3.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.4	9.6	9.4	10.0	11.0	10.5	10.4	10.0	10.8	9.0	8.1	7.7	(35)
(36)			MCD BR	10.5	10.7	12.4	15.6	14.2	12.9	11.9	11.5	12.5	12.5	11.8	8.8	(36)
(37)			MCWBR	9.5	8.9	8.6	9.6	7.6	6.6	5.7	5.8	6.7	7.5	9.0	8.1	(37)
(38)		5% WB	MCD BR	10.1	9.3	10.8	13.2	11.8	10.4	10.0	9.4	11.2	10.6	11.0	8.5	(38)
(39)			MCWBR	9.5	8.1	8.2	9.1	7.5	5.9	5.4	5.2	6.8	7.3	9.3	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.291	0.324	0.358	0.380	0.426	0.432	0.412	0.374	0.332	0.308	0.289	(40)	
(41)		taud	2.377	2.268	2.327	2.364	2.359	2.230	2.220	2.272	2.371	2.494	2.482	2.351	(41)	
(42)		Ebn at Noon	865	888	907	902	888	845	833	836	842	838	797	775	(42)	
(43)		Edh at Noon	73	99	109	115	120	137	137	125	104	80	67	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.44	2.38	3.60	4.34	5.09	5.60	5.64	5.00	3.97	2.29	1.39	1.00	(44)	
(45)		RadStd	0.16	0.25	0.29	0.51	0.49	0.43	0.42	0.23	0.32	0.21	0.18	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+59	N/A

Nomenclature: See separate page